Mehul Agarwal

Software Engineer | AI & LLM Applications | Full-Stack Development

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SUMMARY

Computer Science graduate (2026) and end-to-end builder who ships AI-powered tools, from multi-agent LLM platforms to production data pipelines. Experienced building with LLMs (LangGraph, RAG, MCP, Azure OpenAI), scoping ambiguous problems alongside technical and non-technical stakeholders, and translating complex systems into things teammates can own. Thrives in small teams wearing multiple hats; 700+ DSA problems solved.

EDUCATION

Jaypee Institute of Information Technology — **2022 – 2026**

Bachelor of Technology, Computer Science and Engineering — *Noida, India*

CGPA: 8.2 / 10

WORK EXPERIENCE

Navikenz — **Mar 2026 – Jun 2026**

Software Engineer Intern, Data Engineering and Backend — *Noida, India*

- Engineered ETL ingestion pipelines on Microsoft Fabric, consolidating data from **3 heterogeneous sources** (Snowflake, Amazon S3, SharePoint) into unified target databases, cutting manual data-transfer effort by **60%**.
- Designed source-to-target mappings and Apache Spark transformations across **5+ schemas**, applying validation and data-quality checks that reduced ingestion errors by **30%** and ensured production-grade reliability.
- Scoped requirements with cross-functional engineers and non-technical product stakeholders in an Agile environment, diagnosing pipeline failures and resolving **100%** of reported data-flow incidents within agreed SLAs.

Pristine Business Solutions — **May 2025 – Jul 2025**

Software Developer Intern, Enterprise Systems and Data Validation — *Noida, India*

- Implemented automated data validation pipelines that improved workflow reliability by **35%** across enterprise production systems.
- Optimized complex SQL queries through index tuning and query restructuring, reducing average execution time by **40%** on large datasets.
- Built anomaly-detection monitoring and automated reporting tools, then documented and handed them off to operations staff, increasing data accuracy and speeding decisions.

PROJECTS

NaviformatIQ – AI Multi-Agent Document Reformatting Platform — GitHub Live

- Scoped, built, and deployed end-to-end an AI multi-agent platform with **3 domain-agnostic workflows** (document generation, style transfer, compliance checking) using LangGraph and Azure OpenAI; maps draft content into a template structure via RAG and surfaces a side-by-side diff for human review.
- Designed a human-in-the-loop review flow plus a context-aware chat agent so non-technical users stay in control of every agent decision and can verify output before export.
- Built a FastAPI backend orchestrating multi-agent pipelines with Neon Postgres, Qdrant vector search, and Server-Sent Events for real-time streaming of agent progress; React (Vite + Tailwind) frontend exports final reports to `.docx` and `.pdf`.

Tech Stack: Python, FastAPI, LangGraph, RAG, Azure OpenAI, React, Docker, Qdrant, PostgreSQL

Sovaria – Authority-Driven Backend-as-a-Service Platform — GitHub Live

- Constructed end-to-end a distributed Backend-as-a-Service spanning **11 specialized engines** (identity, policy, NoSQL/SQL storage, temporal versioning, realtime, simulation, functions, billing) behind a single Identity → Policy → Execution pipeline.
- Delivered low-latency WebSocket realtime (5–20 ms from commit) and shipped identical Python and TypeScript SDKs plus a CLI with dry-run simulation for CI/CD, enabling adoption without reading the internals.

Tech Stack: Python, TypeScript, Distributed Systems, NoSQL and SQL Engines, WebSockets, REST API

Code Similarity and Plagiarism Detection System — GitHub Live

- Built a plagiarism-detection tool for educators and academic settings using AST parsing, tokenization, and control-flow analysis to flag structural code similarity across large submission sets.
- Streamlined multi-file processing to compute pairwise similarity scores efficiently, with an interactive dashboard that ranks and highlights suspicious pairs for a human reviewer to make the final call.

Tech Stack: Python, Flask, Algorithms, SQL

LEADERSHIP AND COMMUNITY

- Mentored junior students in Data Structures and Algorithms, breaking down problem-solving patterns and guiding hands-on practice – translating technical concepts for less-experienced peers.
- **Lead Coordinator, Optica Club** (2025): scoped and ran a campus coding contest end-to-end – problem selection, logistics, and prize-money distribution – for student participants.

SKILLS

Languages: Python, TypeScript, C++, SQL, JavaScript

AI/ML and GenAI: LLMs, LangChain, LangGraph, RAG, MCP, Agent Frameworks, Azure OpenAI, Vector Databases, Prompt Engineering, TensorFlow

Web and Backend: FastAPI, Node.js, Express.js, React.js, REST APIs, WebSockets, Server-Sent Events

Data Engineering: ETL/ELT Pipelines, Apache Spark, Microsoft Fabric, Snowflake, Data Modeling, Data Quality, Pandas, NumPy

Cloud and DevOps: Microsoft Azure, AWS (S3), Docker, CI/CD, Git, GitHub

Databases: PostgreSQL, MySQL, MongoDB, Qdrant